

detached summary producers

RLB forward hook

derivative gain $\hat{d}^{\text{live}}$
response gain $\hat{o}^{\text{live}}$

matrix pressure EMAs

$p^{\text{in}}, p^{\text{out}} \rightarrow q^{\text{in}}, q^{\text{out}}$
pressure contrast π

coefficient gradients

$p^R \rightarrow q^R$ activity EMA
activity contrast α

pair state and probes

probe gains $\hat{d}^{\text{probe}}, \hat{o}^{\text{probe}}$
post-step log-ratio γ

sg(detached read-only summary bus)

MatrixPolicy

gate controls

live gains; π ; α

matrix-step controls

role/depth; Muon reads π , α

pair-rescale controls

probe gains, γ

$\pi, \alpha \mapsto \tau, \chi^{\text{rs}}, \ell$

matrix gradients

$\nabla A_{l,g}, \nabla B_{l,g}$

1. gate gradients

scale $\nabla A_{l,g}$
scale $\nabla B_{l,g}$

2. matrix step

AdamW role/depth
early Muon substep

$\Delta A_{l,g}$
 $\Delta B_{l,g}$

3. apply and rescale

add $\Delta A, \Delta B$ to pair

$A_{l,g} \leftarrow e^{\ell} A_{l,g}$
 $B_{l,g} \leftarrow e^{-\ell} B_{l,g}$

updated
 A_l, B_l

current RLB pair

$M_{\text{in}} = A_l$
 $M_{\text{out}} = B_l$

AdamW-only bypass

rational coefficients
and non-RLB tensors

AdamW

AdamW update only
no gates, Muon, or rescale